

HexaHome QA Assessment Report

1. Introduction

As part of this assignment, I tested the HexaHome web application and Android application to understand the product functionality and identify any issues affecting the user experience. The testing mainly focused on property search, filters, authentication, property listings, bookmarks, agents, builders, projects, careers, and profile-related features.

The objective was to evaluate the application's functionality, performance, SEO visibility, and overall usability from an end-user perspective.

2. Performance Testing

Performance testing was conducted using Google Lighthouse on the Home Page, Search Page, and Property Details Page for both desktop and mobile devices.

Observations

During testing, I noticed that desktop pages loaded relatively faster and provided a smoother experience. Mobile performance was comparatively lower, especially on the Search Page and Property Details Page.

The Property Details Page took longer to load on mobile devices because of multiple images and content sections. The Search Page also showed some delay while loading filtered results.

Suggestions

- Compress property images before loading.
- Implement lazy loading for images.
- Reduce unnecessary JavaScript files.
- Improve API response time for property data.
- Enable caching for frequently visited pages.

Overall, the website performance was acceptable, but there is room for improvement on mobile devices.

3. SEO Testing

SEO testing was performed using the provided search queries. A total of 95 keywords related to properties, rentals, apartments, plots, and location-based searches were checked.

Observations

For many keywords, HexaHome pages were visible in Google search results. The website appears to have good visibility for several property-related searches.

However, for some location-specific keywords, rankings were either lower than expected or the relevant page was not easily visible.

Suggestions

- Create more city and locality-specific landing pages.
- Improve page titles and meta descriptions.
- Add FAQ sections to property pages.
- Improve internal linking between related pages.
- Implement structured data for property listings.

4. Functional Testing

Functional testing was performed on major features available on the platform.

Modules Tested

- User Login and Registration
- Property Search
- Property Filters
- Property Details
- Bookmark Feature
- My Listings
- Agents
- Builders
- Projects
- Careers
- User Profile

Test Execution Summary

A total of 80 website test cases were executed.

Out of the major user flows tested:

- 3 flows passed successfully.
- 2 flows failed due to functional issues.

For Android testing:

- 3 flows passed successfully.
- 2 flows showed issues during execution.

Most core features were functional, but issues were found mainly in filtering and result accuracy.

5. Bug Summary

During testing, I identified 16 bugs across different modules.

Major Issues Found

- Budget filter displayed properties outside the selected price range.
- Property condition filter returned incorrect results.
- Property type filter was not working as expected.
- Area filter showed unrelated properties.
- Sorting options did not always display correctly sorted results.
- Some property detail pages showed inconsistent information.

Other Issues

- Facing filter issues.
- "With Photos" filter not working correctly.
- Hot Deals filter inconsistencies.
- Minor UI and responsiveness issues.
- Android application workflow issues.

These issues can affect the property search experience and may reduce user confidence in search results.

6. Bug Fixing Plan

Selected Bug

Budget Filter Displays Incorrect Properties

Issue Description

While testing the search filters, I selected a specific budget range. However, properties outside the selected range were still displayed in the search results.

Possible Cause

The filtering logic may not be correctly validating minimum and maximum budget values before returning search results.

Proposed Solution

- Review backend filter conditions.
- Validate budget range inputs.
- Ensure API responses only return matching properties.

- Add test cases for different budget combinations.

Verification

After implementing the fix, I would retest:

- Minimum budget values
- Maximum budget values
- Boundary values
- Combined filter scenarios

I would also perform regression testing on all property search filters.

7. Security Recommendations

Based on my observations, the following security improvements can help strengthen the application:

- Enforce HTTPS across all pages.
- Limit OTP requests to prevent abuse.
- Encrypt sensitive user information.
- Secure API communication using authentication tokens.
- Validate all user inputs on both client and server sides.
- Implement proper session management.
- Monitor suspicious login activity.

8. Best Practices

To improve maintainability and scalability, I would recommend:

- Using Git for version control.
- Performing code reviews before deployment.
- Implementing CI/CD pipelines.
- Maintaining proper API documentation.
- Automating regression testing where possible.
- Monitoring application logs regularly.
- Optimizing database queries for faster performance.

9. Collaboration Approach

While working in a QA team, I would coordinate closely with developers, designers, and product managers.

My approach would include:

- Understanding requirements before testing.
- Preparing test cases based on user stories.
- Reporting bugs with clear reproduction steps.
- Participating in bug review discussions.
- Retesting fixes before closing defects.
- Providing regular updates on testing progress.

Good communication helps ensure issues are resolved quickly and efficiently.

10. Feedback Handling

I believe feedback is an important part of improving testing quality.

When receiving feedback, I would:

- Review the comments carefully.
- Understand the reason behind the feedback.
- Update test cases if required.
- Improve documentation and bug reporting practices.
- Apply lessons learned in future testing activities.

This helps improve both testing quality and overall team collaboration.

11. Conclusion

Through this assignment, I was able to explore and test various features of the HexaHome platform. The application provides useful functionality for property search and listing management, and most core features worked as expected.

However, several issues were identified in filtering, search accuracy, and mobile workflows. Addressing these issues can improve user experience and overall product reliability.

Overall, the assignment provided valuable insight into the platform and helped demonstrate a structured approach to QA testing and defect reporting.